

EN401 — Handout 4 Results & Analysis Report

Upstream Fuel Supply Options & Economic Merit Order Behavior

1. Solver Metrics & Dimensions

- **Status:** OPTIMAL | **Objective Value (NPV):** \$51,009,521.16 (0.0% delta vs. HO3)
- **Problem Scale:** 872 Rows (+406), 480 Columns (+240), 4,320 Non-Zeros | **Runtime:** < 0.05s

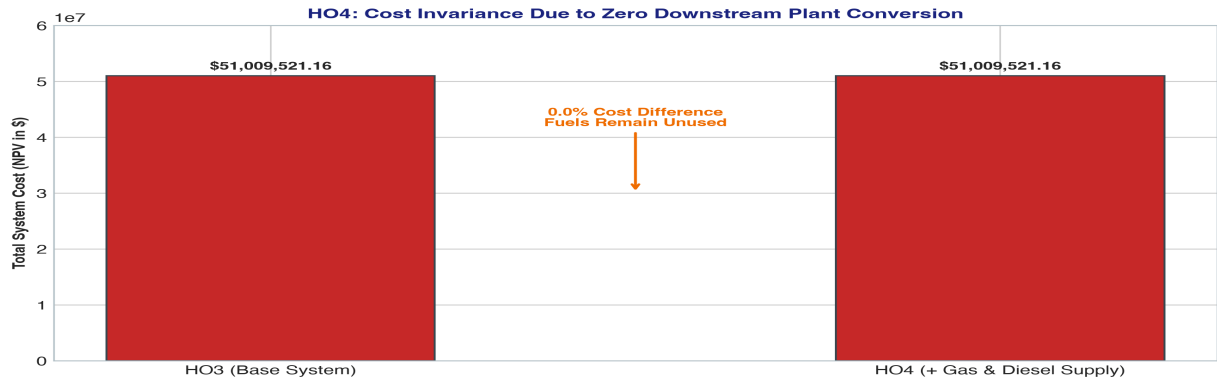


Figure: System cost remains identical due to zero downstream conversion plants.

2. Comparative Cross-Scenario Matrix

Scenario	Active Upstream Fuels	Power Conversion Assets	NPV System Cost	Cost Delta
HO3 (Base)	Penalty Virtual Fuel (`MINBACK`)	Virtual Generator (`BACKSTOP`)	\$51,009,521.16	Baseline
HO4 (Fuels)	`MINNGS` (Gas) + `IMPDSL` (Diesel)	Virtual Generator (`BACKSTOP`)	\$51,009,521.16	0.0% (Unchanged)

3. Engineering Merit Order Takeaway

Introducing fuels without power plants leaves fuels stranded. The LP solver strictly maintains energy conservation, requiring BACKSTOP until power plants arrive in HO5.